SHOPSMART RETAIL LTD (BUSINESS REPORT)

Part 1: Data Understanding

Introduction

The project contains nine CSV datasets and a Data Dictionary Excel file. This review identifies each dataset's purpose, potential primary and foreign keys, and possible relationships before data preparation and analysis.

Dataset Overview

Dataset	Purpose	Potential Primary Keys	Potential foreign keys
Customers.csv	Customer details and segments	Customer _ID	None identified
Products.csv	Product details and prices	Product _ID	None identified
Orders.csv	orders and transaction details	Order _ID	Customer _ID, Product _ID, Store _ID
Stores.csv	Store locations and details	Store _ID	None identified
Returns.csv	Returned Orders	Return_ ID	Order_ ID
Employees.csv	Employee details	Employee_ ID	Store_ ID
Suppliers.csv	Supplier details	Supplier_ ID	None identified
Inventory.csv	stock and restocking details	Inventory_ ID	Product _ID, Store _ID
Order_ items.csv	Items within orders	Order_ Item_ ID	Order_ ID, Product _ID

Primary keys are candidate keys based on initial uniqueness checks. Foreign keys and relationships are proposed from the identified columns and require validation.

Possible Relationships
Customers → Orders (Customer_ ID)

Stores → Orders, Employees and Inventory (Store_ ID)

Orders → Returns and Order_ Items (Order_ ID)

Products → Inventory and Order_ Items (Product_ ID)

No direct relationship for Suppliers has been identified.

Conclusion: The review provides an initial understanding of the datasets and their potential connections for subsequent data preparation and sales analysis

Part 2: Data Cleaning Summary

Tool Used: Microsoft Excel

Objective

The objective of data cleaning was to check the nine supplied datasets for common data-quality issues and ensure they were organized and suitable for analysis.

Data Cleaning Activities Performed

1. Duplicate Checks

The datasets were checked for duplicate records. No exact duplicate rows were found during the checks.

2. Missing Values

Excel's Go To Special → Blanks feature was used to check for empty cells. No blank cells were identified during these checks.

3. Data Formatting

Date columns were reviewed to ensure dates were displayed consistently. Currency and numerical fields were also checked to ensure their formats were appropriate for analysis.

4. Phone Number Consistency

Inconsistent phone-number formats were identified in the Customers dataset, including different formats and extensions. These values were noted for attention rather than changed without verification.

5. Table Organization

The datasets were reviewed to ensure they had appropriate column headers and were converted into Excel Tables to support structured references and subsequent analysis.

Outcome

The data-cleaning process helped identify potential data-quality issues and prepare the datasets for further analysis. No exact duplicate rows or blank cells were found during the checks performed. Inconsistent phone-number formats were identified as an issue requiring careful handling.

The cleaned and organized datasets were then used for the subsequent stages of the Shop Smart Retail Analytics project.

BUSINESS QUESTIONS AND FINDINGS

Introduction

This section presents the findings from the analysis of Shop Smart's retail data using Microsoft Excel. The analysis addresses ten business questions relating to product revenue, store performance, product categories, monthly sales, customer spending, product returns, supplier information, inventory levels, regional performance, and investment opportunities.

Question 1: Which product generated the highest revenue?

Beauty Item 210 generated the highest revenue, with total sales of 8,075,557.00 from completed orders. This indicates that the product was the strongest revenue contributor among the products analyzed. Management should consider maintaining sufficient stock and monitoring customer demand to sustain its sales performance.

Question 2: Which store performed best?

Shop Smart Lagos recorded the highest total sales, amounting to 419,596,291.00 from completed orders. This makes it the leading store by revenue among the stores analyzed. Management should investigate the factors contributing to its strong performance and consider applying successful practices to other stores.

Question 3: Which category sells the most?

The Beauty category recorded the highest total sales, amounting to 581,030,145.00 from completed orders. This indicates that Beauty was the leading product category by revenue among the categories analyzed. Management should continue monitoring demand in this category and ensure that popular beauty products remain adequately stocked.

Question 4: Which month generated the highest revenue?

December recorded the highest total sales, amounting to 224,722,471.00 from completed orders. This suggests that December was the strongest sales month among the months analyzed. Management should investigate the factors contributing to this performance and prepare inventory and staffing to meet potential increases in customer demand during this period.

Question 5: Which customers spend the most?

Nicole Smith recorded the highest total sales, amounting to 3,812,964.00 from completed orders. This makes Nicole Smith the highest-revenue-generating customer among those

analyzed. Management should study the purchasing patterns of high-value customers and consider targeted loyalty programs to encourage repeat purchases and strengthen customer retention.

Question 6: Which products are frequently returned?

Beauty Item 6-1 had the highest count of returned IDs, with seven records in the analysis. This indicates that it was the product most frequently associated with returned orders in the results analyzed. Management should investigate the possible reasons for these returns and review product quality, descriptions, and customer feedback to identify opportunities for improvement.

Note: The Returns dataset does not directly identify the specific product returned. Therefore, this finding is based on the association between products and returned orders and does not confirm that all seven return records relate directly to Beauty Item 6-1.

Question 7: Which supplier supplies the highest number of products?

The available datasets did not provide an identifiable relationship between the Suppliers table and the Products table. Although supplier information was provided, no matching supplier key was identified in the other datasets to establish which products each supplier supplies.

Therefore, it was not possible to determine which supplier supplies the highest number of products from the available data. To perform this analysis, Shop Smart would need a reliable supplier identifier linking each supplier to the products they supply.

Question 8: Which stores have low inventory?

Shop Smart Kaduna (Store ID: STR011) recorded the lowest total stock on hand, with 2,570 units. This suggests that the store may have lower stock availability than the other stores analyzed. Management should review the store's inventory requirements, monitor demand, and consider replenishing stock where necessary to reduce the risk of stockouts.

Note: This finding is based on total stock quantity, not inventory value.

Question 9: Which region performs best?

The East region recorded the highest total sales, amounting to 694,249,970.00 from completed orders. This makes the East the best-performing region by revenue among the regions analyzed. Management should investigate the factors contributing to the region's strong performance and consider applying successful strategies to other regions to improve overall sales.

Question 10: Which product category should management invest in?

The Beauty category recorded the highest total sales, amounting to 581,030,145.00 from completed orders. Based on its strong revenue performance, management should consider investing further in the Beauty category by maintaining adequate stock, identifying popular products, and improving marketing efforts.

However, before increasing investment, management should also review the category's profitability, inventory costs, and customer demand to ensure that higher sales translate into sustainable profits.

Summary of Findings

The analysis identified Beauty Item 210 as the highest-revenue-generating product, Shop Smart Lagos as the best-performing store, and Beauty as the leading product category by sales. December recorded the highest monthly sales, while Nicole Smith was the highest-revenue-generating customer. Beauty Item 6-1 had the highest count of returned IDs in the analysis, although the available Returns dataset does not directly identify the specific products returned. Shop Smart Kaduna had the lowest total stock quantity, while the East region recorded the highest total sales. The supplier analysis could not be completed because a reliable link between suppliers and products was not available.

Overall, these findings provide useful insights into Shop Smart's sales performance, customer activity, product returns, regional performance, and inventory levels. They can help management identify areas of strength and opportunities for further investigation and improvement.

BUSINESS RECOMMENDATIONS

Shop Smart Retail Analytics Project

1. Three Key Findings

1. Product Category Performance: The Beauty category generated the highest total sales of 581,030,145.00, making it the leading category by revenue in the analysis.

2. Store and Regional Performance: Shop Smart Lagos recorded the highest store sales of 419,596,291.00, while the East region recorded the highest regional sales of 694,249,970.00.

3. Inventory and Returns: Shop Smart Kaduna had the lowest total stock on hand, with 2,570 units. Beauty Item 6-1 had the highest count of returned IDs associated with products in the analysis, with seven records. The returns data does not directly identify the specific products returned.

2. Three Business Recommendations

1. Invest Strategically in Beauty Products: Management should review demand, profit margins, and inventory costs before investing further in the Beauty category. Popular products should be adequately stocked to meet customer demand.

2. Improve Store and Regional Performance: Management should investigate the factors behind the strong sales performance of Shop Smart Lagos and the East region. Successful strategies should be assessed for application in other stores and regions.

3. Strengthen Inventory and Returns Management: Management should monitor stock levels across stores and replenish inventory where necessary. The business should also improve return records by linking each return to the specific product and investigating recurring return patterns.

3. One Risk Identified

Data Quality and Incomplete Relationships: The available datasets do not establish a reliable relationship between suppliers and products, and the Returns dataset does not directly identify returned products. These limitations may affect the accuracy and completeness of supplier and product-return analysis, potentially leading to less informed business decisions.

4. One Opportunity Identified

Growth in the Beauty Category: The Beauty category's leading sales performance presents an opportunity for Shop Smart to strengthen its product selection, improve targeted marketing, and maintain appropriate stock levels. Reviewing profitability alongside revenue can help management identify sustainable growth opportunities.

Conclusion

The analysis highlights opportunities to build on strong sales performance while improving inventory control and data quality. By making evidence-based investment decisions, learning from high-performing stores and regions, and improving operational records, Shop Smart can strengthen its retail performance and business decision-making.